

Word problems on percentages

Shops, salaries, populations and exam marks — all the same three questions in disguise.

LESSON 67–70

4 × 40 MIN

MATEMATIKA 6, §15

S7 10

LESSON CLOCK

30'

40'

50'

30'

10

Naming the question	30 min
Shops and money	40 min
Successive changes	50 min
Populations and marks	30 min
Homework	10 min

LEARNING OBJECTIVES

- Choose which percentage question a problem is asking.
- Work with successive percentage changes.
- Solve problems about discounts, taxes and interest.
- Interpret a percentage answer in the context it came from.

TEXTBOOK

National	pp. 191–202
Cambridge	Stage 7, pp. 96–104
Workbook	Exercise 10.4

01

Explanation

Naming the question

WORDING	QUESTION	OPERATION
“find 15% of...”	a part	multiply
“...is 15% of what?”	the whole	divide
“what percentage of... is...?”	the percentage	$\frac{\text{part}}{\text{whole}} \cdot 100$
“increase by 15%”	a new value	$\cdot 1.15$
“after a rise of 15% it is...”	the original	$\div 1.15$

FIVE PHRASES, FIVE OPERATIONS

Nearly every percentage problem in the paper opens with one of those five wordings. Matching the phrase to the operation is more reliable than reasoning from the numbers.

Shops and money

PROBLEM	WORKING	ANSWER
a jacket at 400 000 with 25% off	$\cdot 0.75$	300 000
a bill of 80 000 plus 15% service	$\cdot 1.15$	92 000
a salary of 2 400 000 rising 8%	$\cdot 1.08$	2 592 000
1 000 000 at 14% simple interest for a year	$\cdot 0.14$	140 000 interest
a sale price of 300 000 after 25% off	$\div 0.75$	400 000

THE LAST ROW IS THE FIRST ROW BACKWARDS

Shops advertise the discount and the sale price; finding the original from them is the question most likely to appear in an examination.

Successive changes

Two changes multiply — they never add.

CHANGES	COMBINED MULTIPLIER	NET CHANGE
up 20%, then up 10%	$1.2 \cdot 1.1 = 1.32$	+32%
up 20%, then down 20%	$1.2 \cdot 0.8 = 0.96$	−4%
down 10% twice	$0.9 \cdot 0.9 = 0.81$	−19%
up 50%, then down 50%	$1.5 \cdot 0.5 = 0.75$	−25%

+20% THEN +10% IS NOT +30%

It is +32%, because the second rise is taken from the already-increased amount. Adding percentage changes is the commonest error in this whole topic.

Populations and marks

PROBLEM	WORKING	ANSWER
a village of 2 500 grows 4%	$\cdot 1.04$	2 600
a mark of 34 out of 40	$\frac{34}{40} \cdot 100$	85%
18 of 24 pupils passed	$\frac{18}{24} \cdot 100$	75%
attendance rose from 80% to 92%	$92 - 80$	12 percentage points

PERCENTAGE POINTS ARE NOT PERCENTAGES

A rise from 80% to 92% is 12 percentage points — but a 15% increase in the attendance figure. Newspapers confuse the two constantly; a mathematician should not.

02

Worked examples

EXAMPLE 1 A jacket costing 400 000 sum is reduced by 25%. Find the sale price.

1

A fall of 25% means multiplying by 0.75.

2

$400\,000 \cdot 0.75$

3

$= 300\,000$ sum.

ANSWER

300 000 sum

EXAMPLE 2 A price rises 20% and then falls 20%. What is the net change?

- ① Multipliers: 1.2 and 0.8.
- ② $1.2 \cdot 0.8 = 0.96$
- ③ That is 96% of the original — a fall of 4%.

ANSWER

−4%

EXAMPLE 3 A pupil scores 34 out of 40. Give the mark as a percentage.

- ① $\frac{34}{40}$
- ② $= 0.85$
- ③ 85%

ANSWER

85%

03 Interactive model

Collect real advertisements with percentages on them; the class checks the arithmetic and often finds the offer is worse than it looks.

Which question, and what answer?

400 000 with 25% off:

100 000

300 000

375 000

500 000

A bill of 80 000 plus 15%:

92 000

95 000

12 000

68 000

A sale price of 300 000 after 25% off; the original:

375 000

400 000

325 000

225 000

Up 20% then up 10% is:

+30%

+32%

+2%

+22%

Up 20% then down 20% is:

0%

-4%

+4%

-40%

Down 10% twice is:

-20%

-19%

-21%

-100%

34 out of 40 as a percentage:

80%

85%

75%

88%

From 80% to 92% is:

12 percentage points

12% increase

both the same

92%

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Discount	Chegirma	Скидка
Tax	Soliq	Налог
Interest	Foiz (bank)	Проценты
Salary	Maosh	Зарплата
Population	Aholi	Население
Successive change	Ketma-ket o'zgarish	Последовательное изменение
Net change	Yakuniy o'zgarish	Итоговое изменение
To interpret	Talqin qilmoq	Истолковать

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

“Increase by 15%” means multiply by:

0.15

1.15

15

0.85

“After a rise of 15% it is...” means:

multiply by 1.15

divide by 1.15

subtract 15%

add 15%

Two successive changes:

add

multiply

subtract

cancel

A salary of 2 400 000 rising 8% becomes:

2 592 000

2 592

2 400 008

2 208 000

1 000 000 at 14% for a year earns:

14 000

140 000

1 400 000

114 000

A rise from 80% to 92% is an increase of:

12%

15%

92%

8%

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. 400 000 with 25% off

ANSWER 300 000

2. 80 000 plus 15%

ANSWER 92 000

3. 2 400 000 up 8%

ANSWER 2 592 000

4. 1 000 000 at 14% for a year: the interest

ANSWER 140 000

5. 2 500 up 4%

ANSWER 2 600

6. 34 out of 40

ANSWER 85%

7. 18 out of 24

ANSWER 75%

Medium · the standard question

7 PROBLEMS

1. A sale price of 300 000 after 25% off: the original

ANSWER 400 000

2. Up 20% then up 10%

ANSWER +32%

3. Up 20% then down 20%

ANSWER -4%

4. Down 10% twice

ANSWER -19%

5. Up 50% then down 50%

ANSWER -25%

6. From 80% to 92%: the increase

ANSWER 15%

7. A price of 92 000 after 15% was added: before

ANSWER 80 000

Hard · stretch and reason

7 PROBLEMS

1. A shop adds 20% then advertises 20% off: the net change

ANSWER -4%

2. A population of 2 500 after two years at 4% a year

ANSWER 2 704

3. A price falls 30% to 84 000: the original

ANSWER 120 000

4. Which is better: 30% off, or 20% off then 10% more off?

ANSWER 30% off — the other leaves 72%

5. A mark rose from 60 to 75 out of 100: the percentage increase

ANSWER 25%

6. A salary rises 10% then falls 10%: the net change

ANSWER -1%

7. Why do successive percentages multiply?

ANSWER Each acts on the amount left by the one before

07 Homework

Homework — 5 tasks

Write the multiplier for every change before you touch the numbers.

1. A coat costs 520 000 sum and is reduced by 35%. Find the sale price.
2. A bill of 140 000 has 12% added. Find the total.
3. After a discount of 40% a price is 96 000. Find the original price.
4. A price rises 25% and then falls 20%. Find the net change.
5. A pupil scores 57 out of 60. Give the mark as a percentage.